Chapter 1 Database Fundamentals

A database is an organised collection of structured data that supports efficient storage, retrieval and modification. A database management system (DBMS) manages access,
enforces constraints and handles concurrent users.

The relational model stores data in tables made of rows (records) and columns (attributes). A primary key uniquely identifies each row. A foreign key is a column that references the primary key of another table and enforces referential integrity.

SQL is the standard language for defining schemas and querying relational databases.

Chapter 2 SQL Joins

A join combines rows from two or more tables using a related column. An INNER JOIN returns only rows where the join condition matches in both tables. A LEFT JOIN returns every row from the left table plus matching right-table rows, filling missing values with NULL.

A FULL OUTER JOIN returns all rows from both tables. A CROSS JOIN produces the cartesian
product: every left row paired with every right row, which is rarely useful without a filter.

A self join joins a table to itself, useful for hierarchies such as employees and managers.

Chapter 3 Normalization

Normalization organises relational data to reduce redundancy and prevent update, insert and delete anomalies.

First normal form (1NF) requires atomic column values. Second normal form (2NF) removes partial dependencies on a composite key. Third normal form (3NF) removes transitive dependencies, so non-key attributes depend only on the primary key.

Boyce-Codd normal form (BCNF) requires every determinant to be a candidate key. Denormalization intentionally reintroduces redundancy to speed up read-heavy workloads.

Chapter 4 Indexing and Query Performance

An index is an auxiliary structure that speeds up lookups at the cost of storage and slower writes. A B-tree index supports equality lookups and range queries efficiently, which makes it the default index type. A hash index supports fast equality lookups but not range queries.

A covering index includes all columns required by a query, so the database can answer from the index without reading table rows. Composite indexes are order sensitive.

EXPLAIN shows whether a query uses an index scan or a sequential scan. Add indexes for measured slow queries, because every index increases write cost.